DetGPT: In-Context Object Detection via Visual Example Prompting on VLMs

02501 Advanced Deep Learning in Computer Vision

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17 March, 2026

Keywords/lectures: VLMs, In-Context Learning, Few-Shot Object Detection, Visual Prompting

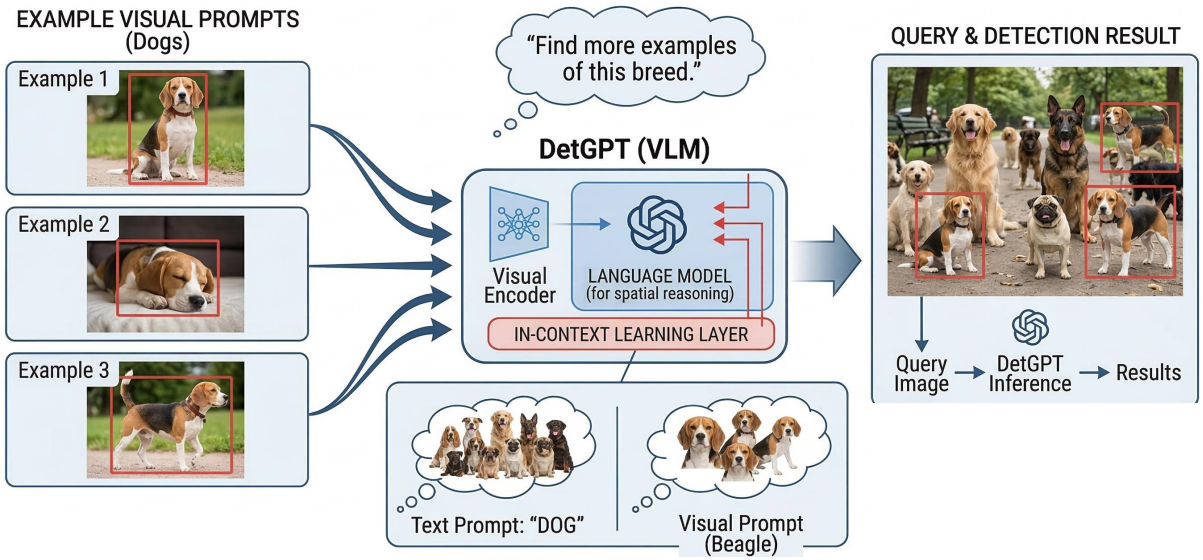

Figure 1: **DetGPT.** Examine in-context learning on VLMs for improving capabilities on localizing objects (object detection)

1 Project description

Few-shot object detection, i.e., detecting a novel object category given only 1–5 annotated examples, is a critical problem in computer vision. In real-world applications (factory defect inspection, wildlife monitoring, medical imaging, autonomous driving in new cities), collecting thousands of labeled bounding boxes for every new object class is impractical. Traditional few-shot detection methods [1, 2] rely on meta-learning architectures trained on base classes and transferred to novel ones. But a recent studies [3] showed that these approaches are outperformed by a much simpler baseline based on zero-shot inference with VLM-based detectors.

However, zero-shot VLM detectors have a fundamental limitation: they define the target class with text only ("detect all trucks"). Text prompts are ambiguous and cannot capture the precise visual concept the user has in mind. Instead of describing the target in words, the alternative is to show the model a few annotated example images. This is in-context learning (ICL) for object detection. This is the visual analog of how GPT learns tasks from examples in its prompt.

Can off-the-shelf VLMs (Qwen2.5-VL, InternVL, GPT-4o) perform few-shot object detection when shown visual examples of the target object with bounding boxes, without any fine-tuning? Modern VLMs accept multi-image inputs, so you can literally show them "here are 3 images with this object annotated with red boxes. Now find all instances of this object in the new image."

2 Data

Use standard few-shot object detection benchmarks: COCO (80 categories, standard base/novel splits: 60 base, 20 novel, with 1/5/10-shot evaluation) or LVIS (1203 categories, many rare categories with very few training examples). For the in-context examples, construct support sets by sampling K images (K=1, 3, 5) containing annotated instances of the target class, rendering bounding boxes as colored overlays (red boxes, numbered labels). Use Qwen2.5-VL-7B [7] and InternVL-2.5-8B [8] as the primary VLMs (multi-image capable, fit on 1 GPU).

If you need detection modules, you can consider Grounding DINO [4] and YOLO-World [5]. For traditional FSOD baselines, you could use DINOv2-based prototype matching and FM-FSOD [10] (foundation model few-shot detection).

3 Tasks

In this project, you could work on the following tasks:

Task 1: Zero-shot performance. Evaluate the zero-shot performance of VLMs on the task of object detection. Smaller VLMs (1-8B parameters) often fail to perform well and on par with SOTA object detectors (DETR-based, Grounding DINO, YOLO, etc..)

Task 2: In-Context Detection Prompt Design. Design and evaluate different strategies for presenting annotated support examples to VLMs for detection: (a) Side-by-side layout, (b) Cropped exemplars, (c) Text-from-vision (d) Set-of-Mark + visual examples. Evaluate all strategies in 1-shot and 5-shot settings on novel classes. Compare with zero-shot Grounding DINO (text prompt only) (ii) zero-shot VLM (text prompt only), (iii) traditional FSOD methods [1, 2].

Task 3: VLM + Detector Fusion Pipeline. VLMs excel at understanding what the target object is from visual examples,but are poor at precise localization. Specialized detectors are precise but need the right text prompt. Combine both: (a) VLM-to-text pipeline: show the VLM the support examples and ask it to generate the optimal text prompt for Grounding DINO (including detailed description, visual attributes, typical context). (b) VLM-as-verifier: run Grounding DINO with a broad text query, then show each detection (cropped) to the VLM alongside the support examples and ask "does this detection match the target object?" for filtering false positives. (c) VLM-guided NMS: when Grounding DINO produces overlapping detections, show them to the VLM to select the best one. Evaluate mAP improvement over Grounding DINO alone.

Task 4: ...

Task 5: ...

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